

Wireless Networking

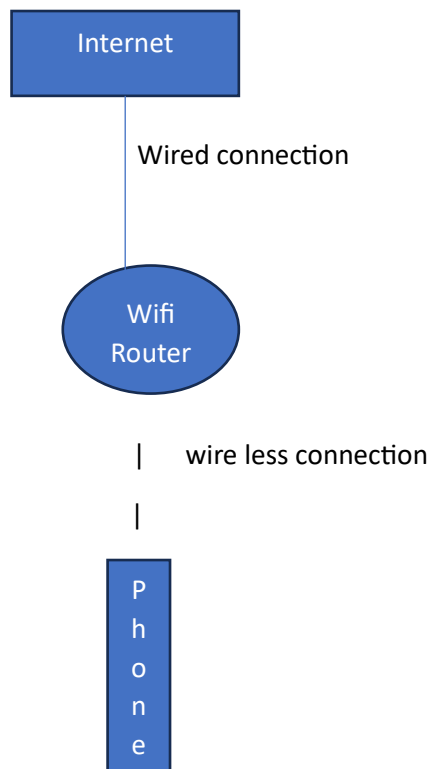
1. List 3 popular Wi-Fi standards (e.g., 802.11n, 802.11ac, 802.11ax) and for each, write one key difference in speed or frequency band.

Wi-Fi Standards

Wi-Fi Standard	Key Difference
802.11n (Wi-Fi 4)	Works on 2.4 GHz and 5 GHz bands and supports speeds up to 600 Mbps.
802.11ac (Wi-Fi 5)	Mainly works on the 5 GHz band and provides speeds up to 6.9 Gbps.
802.11ax (Wi-Fi 6)	Works on 2.4 GHz and 5 GHz bands and provides higher speed and better performance in crowded networks.

Conclusion: Each newer Wi-Fi standard generally provides better speed, efficiency, and network performance than the previous generation.

2. Draw a simple diagram (on paper or using any drawing tool) showing how your smartphone connects to the internet via a Wi-Fi router at home, labeling the wireless and wired parts of the network.



3. Scan for available Wi-Fi networks using your laptop or phone, note down the SSID, signal strength, and security type (WPA2, WPA3, etc.) for at least 3 networks, and explain which one you would choose and why.

Wi-Fi Network Scan

I checked the available Wi-Fi networks using my laptop and recorded the following details:

No.	SSID	Signal Strength	Security Type
1	Home_WiFi	Excellent (90%)	WPA2
2	JioFiber_5G	Good (75%)	WPA3
3	TP-Link_5G	Fair (55%)	WPA2

Choice: I would choose **JioFiber_5G** because it has a good signal strength and uses **WPA3 security**, which provides stronger security than WPA2.

4. Explain in your own words what 'SSID hiding' is in Wi-Fi networks and discuss one advantage and one disadvantage of hiding your Wi-Fi SSID.

SSID Hiding

SSID hiding means making the name of a Wi-Fi network invisible from the normal list of available Wi-Fi networks. Users need to manually enter the Wi-Fi name to connect.

- **Advantage:** It can reduce unwanted connection attempts because the network name is not openly visible.
- **Disadvantage:** It does **not provide strong security**, because the hidden SSID can still be discovered using network scanning tools and may make connecting devices less convenient.

Conclusion: SSID hiding can provide a small amount of privacy, but it should not be considered a replacement for strong Wi-Fi security such as WPA2 or WPA3.